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COMPE470

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# Assignment 6

## Module

```
module MEM #(parameter WL = 4, N = 4, DEPTH = 2**(N+1), ADDR_WL = N + 1)
(input CLK, RST, output signed [WL - 1: 0] y_out);
```

```
    reg REN = 1, WEN = 1;
    reg [ADDR_WL - 1: 0] raddr1, raddr2, waddr;
    wire signed [WL - 1: 0] wdata;
    wire signed [WL - 1: 0] rdata1, rdata2;
```

```
    assign wdata = (rdata1 + rdata2) >>> 1;
    assign y_out = wdata;
```

```
    regfile #(.WL(WL), .ADDR_WL(ADDR_WL))
        rf (.CLK(CLK), .REN(REN), .WEN(WEN),
            .raddr1(raddr1), .raddr2(raddr2), .waddr(waddr),
            .wdata(wdata), .rdata1(rdata1), .rdata2(rdata2));
```

```
    initial $readmemb("data.dat", rf.MEM, 0, (2**N)-1);
```

```
    always @(posedge CLK) begin
        if (RST) begin
            raddr1 <= 0;
            raddr2 <= (2**N) - 1;
            waddr <= 2**N;
        end
        else begin
            if (raddr1 <= (2**N)-1) begin
                raddr1 <= raddr1 + 1;
                raddr2 <= raddr2 - 1;
                waddr <= waddr + 1;
            end
            else begin
```

```
        raddr1 <= 0;
        raddr2 <= (2**N) - 1;
        waddr <= 2**N;
    end
end
end
endmodule
```